Geometry

Independent diagnostic, two proof questions

Time allowed: 55 minutes **Total: 14 marks**

Candidate identifier: _____ **Date:** _____

Instructions. Attempt both questions independently. Show complete arguments, including all relevant cases and any final verification. No internet, calculator, AI tools or hints during the timed sitting. If unfinished, write down any useful progress. Stop the timed work when time is called.

Question 1 **[7 marks]**

In an acute triangle ABC, let D, E and F be the feet of the perpendiculars from A, B and C to BC, CA and AB, respectively. Prove that

$$\angle EDF = 180^\circ - 2\angle BAC.$$

A clear labelled diagram may be used, but all claimed angle relationships must be justified.

Question 2 **[7 marks]**

Points A, B, C, D lie on a circle in that order, and form a convex quadrilateral. Without quoting Ptolemy's theorem, prove that

$$AC \cdot BD = AB \cdot CD + BC \cdot AD.$$

You may introduce auxiliary points or lines. Justify any triangles you claim are similar.

After the sitting (not part of the timed assessment): Record approximate time spent on each question, whether either problem was familiar, and whether you worked beyond the limit. If you need more space, use the separate answer sheets.

Q1: _____ min Q2: _____ min Prior exposure: _____